

Erdős Problem 252

Factorial tails, exact cancellation, and one surviving shift

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The theorem concerns every divisor-power sum

For $n \geq 1$ and $k \in \mathbb{N} = \{0, 1, 2, \dots\}$, define

$$\sigma_k(n) = \sum_{d|n} d^k, \quad \alpha_k = \sum_{n \geq 1} \frac{\sigma_k(n)}{n!}.$$

For example, the divisors of 6 are 1, 2, 3, 6:

$$\sigma_0(6) = 4, \quad \sigma_1(6) = 12, \quad \sigma_2(6) = 50.$$

The series converges: $\sigma_k(n) \leq n^{k+1}$, and factorial growth wins.

Main theorem

α_k is irrational for every natural number k .

We explain the positive degrees first; $k = 0$ has a shorter ending.

Earlier results reached the fourth power

Exponent	Prior results
$k = 1, 2$	Erdős–Kac; elementary tail arguments.
$k = 3$	Schlage-Puchta (2006) and Friedlander–Luca–Stoiciu (2007), independently, using sieve methods.
$k = 4$	Pratt (2022), using sieve methods and exponential sums.

Earlier all-degree results assumed Schinzel's hypothesis H or a suitable prime-tuples conjecture.

The argument here uses fixed coprime integers and exact cancellation.

Its only prime-selection step is to choose one sufficiently large prime.

Source: Pratt (2022), introduction.

Warm-up: factorials expose a rational denominator

Suppose $e = \sum_{m \geq 0} 1/m! = a/b$ with $a, b \geq 1$ integers. Then

$$b! e - b! \sum_{m \leq b} \frac{1}{m!} = b! \sum_{m > b} \frac{1}{m!} \in \mathbb{Z}.$$

Yet

$$0 < b! \sum_{m > b} \frac{1}{m!} < \sum_{j \geq 1} \frac{1}{(b+1)^j} = \frac{1}{b} \leq 1.$$

There is no integer strictly between 0 and 1.

The principle we will reuse

A sequence of integers tending to 0 is eventually exactly 0.

Rationality would make all late tails integers

Fix k . Scale the tail after the n -th term:

$$T_n := n! \sum_{m>n} \frac{\sigma_k(m)}{m!}.$$

If $\alpha_k = a/b$ with $a \in \mathbb{Z}$ and $b \geq 1$, then for $n \geq b$,

$$T_n = \underbrace{\frac{n!a}{b}}_{\text{integer}} - \underbrace{\sum_{m=1}^n \sigma_k(m) \frac{n!}{m!}}_{\text{integer}}.$$

All later work begins with this consequence:

$$\alpha_k \in \mathbb{Q} \implies T_n \in \mathbb{Z} \text{ for all sufficiently large } n.$$

The original tails need not become small

The first terms are

$$T_n = \frac{\sigma_k(n+1)}{n+1} + \frac{\sigma_k(n+2)}{(n+1)(n+2)} + \cdots.$$

The divisor $n+1$ alone gives the exact lower bound

$$T_n \geq (n+1)^{k-1} \quad (k \geq 1).$$

- For $k = 1$, the tail is at least 1; it cannot tend to 0.
- For $k \geq 2$, this lower bound tends to infinity.

We need a small integer combination of tails, not a small individual tail.

The proof has four main moves

1. **Expand.** Separate a finite main term from an error smaller than $1/n$.
2. **Align.** Read several tails at compatible dilated indices.
3. **Cancel.** Finite differences remove all lower powers. The weighted integer becomes zero, forcing a remaining sum to tend to zero.
4. **Separate.** Two progression averages detect one surviving shift differently. They cannot both be zero.

Factorials provide integrality; algebra cancels; averages give the contradiction.

1. Reciprocal products have integer coefficients

Center each denominator at its largest factor x :

$$\begin{aligned}\frac{1}{x(x-1)} &= \frac{1}{x^2} + \frac{1}{x^3} + \frac{1}{x^4} + \cdots, \\ \frac{1}{x(x-1)(x-2)} &= \frac{1}{x^3} + \frac{3}{x^4} + \frac{7}{x^5} + \cdots.\end{aligned}$$

In general, for $x > j$,

$$\frac{1}{x(x-1)\cdots(x-j)} = \sum_{i \geq j} \left\{ \begin{matrix} i \\ j \end{matrix} \right\} \frac{1}{x^{i+1}}.$$

$\left\{ \begin{matrix} i \\ j \end{matrix} \right\}$ counts partitions of i objects into j nonempty blocks. In particular, $\left\{ \begin{matrix} i \\ j \end{matrix} \right\} = 0$ for $i < j$, and $\left\{ \begin{matrix} j \\ j \end{matrix} \right\} = 1$.

We keep only the powers through $x^{-(k+1)}$.

1. The discarded part is smaller than $1/n$

With j indexing the shifted argument and $i + 1$ the denominator power,

$$T_n = \sum_{j=0}^k \sum_{i=j}^k \left\{ \begin{matrix} i \\ j \end{matrix} \right\} \frac{\sigma_k(n+j+1)}{(n+j+1)^{i+1}} + E_n.$$

The error has two parts:

- the terms after the first $k + 1$ terms of the actual tail;
- the higher reciprocal powers of each retained denominator.

Divisor pairing gives $\sigma_k(m) \leq 64m^k \sqrt{m}$. A geometric majorant and the finite Stirling remainder bound then give

$$0 \leq E_n \leq C_k n^{-3/2} \quad \text{for large } n, \quad nE_n \longrightarrow 0.$$

Here C_k is a fixed constant depending only on k .

2. Multiplicativity turns dilations into shifts

Choose a positive integer p coprime to $1, \dots, k+1$, and an offset $0 \leq t < p$. For large $N \equiv t \pmod{p^2}$, put $n = (N - t)/p$.

$$p \mid n, \quad \gcd(p, n + j + 1) = \gcd(p, j + 1) = 1 \quad (0 \leq j \leq k).$$

Define $r = (j + 1)p - t$. Then $p(n + j + 1) = N + r$, so

$$\sigma_k(p) \sigma_k(n + j + 1) = \sigma_k(N + r).$$

Consequently,

$$\sigma_k(p) \frac{\sigma_k(n + j + 1)}{(n + j + 1)^{i+1}} = p^{i+1} \frac{\sigma_k(N + r)}{(N + r)^{i+1}}.$$

The multiplier now appears as a polynomial factor p^{i+1} .

2. Example: three tails share the same first shift

For $k = 1$, use three pairwise coprime multipliers, including the composite 9:

	first tail	second tail	third tail
p	5	7	9
t	0	2	4
$n_p = (N - t)/p$	$N/5$	$(N - 2)/7$	$(N - 4)/9$
$p(n_p + 1)$	$N + 5$	$N + 5$	$N + 5$
$p(n_p + 2)$	$N + 10$	$N + 12$	$N + 14$

Choose $N \equiv 0 \pmod{25}$, $N \equiv 2 \pmod{49}$, $N \equiv 4 \pmod{81}$. The Chinese remainder theorem makes these conditions compatible.

The first arguments coincide. The second arguments separate.

3. Finite differences cancel polynomial factors

For every polynomial P of degree at most k ,

$$\sum_{a=0}^{k+1} b(a)P(a) = 0, \quad b(a) = (-1)^{k+1-a} \binom{k+1}{a}.$$

For $k = 1$, this is the second-difference identity:

$$P(0) - 2P(1) + P(2) = 0.$$

In our example $p = 5 + 2a$, so

$$\underbrace{5 - 2 \cdot 7 + 9}_0 \frac{\sigma_1(N+5)}{N+5} = 0.$$

A shared arithmetic value can be arbitrary: its polynomial multiplier cancels exactly.

3. Build one free coordinate for each lower shift

Let $e = (e_0, \dots, e_{k-1}) \in \{0, \dots, k+1\}^k$. Set

$$G = (k+2)^k - 1, \quad D = G!, \quad B = 1 + kDG,$$

and define

$$p_e = B + D \sum_{a < k} e_a (k+2)^a, \quad t_e = D \sum_{a < k} (a+1) e_a (k+2)^a.$$

Their shift at position j is

$$r_e(j) = (j+1)p_e - t_e = (j+1)B + D \sum_{a < k} (j-a) e_a (k+2)^a.$$

For $j < k$, the shift ignores e_j : its coefficient is zero.

But p_e is affine in e_j , so p_e^{j+1} is a polynomial in that coordinate.

3. The grid also guarantees the needed coprimality

Write $I_e = \sum_{a < k} e_a(k+2)^a$. Distinct vertices give distinct integers $I_e \in [0, G]$, and $p_e = B + DI_e \equiv 1 \pmod{D}$.

- **Pairwise coprime.** A common prime divisor ℓ of p_e, p_f would divide $|I_e - I_f| \in [1, G]$, but not D . Then $\ell \leq G$ forces $\ell \mid G! = D$, a contradiction.
- **Coprime to small shifts.** For $k \geq 1$, $k+1 \leq G$; hence every $1, \dots, k+1$ divides D .
- **Positive arguments.** $t_e \leq kDI_e \leq kDG = B - 1 < p_e$, so every $r_e(j) > 0$.

No multiplier needs to be prime, and no prime-pattern conjecture is used.

3. A common progression preserves integrality

Use the product weights $w_e = \prod_{a < k} b(e_a) \in \mathbb{Z}$. Since the p_e^2 are pairwise coprime, choose

$$A \equiv t_e \pmod{p_e^2} \quad \text{for every } e, \quad Q = \prod_e p_e^2.$$

Every $N = A + Qu$ satisfies the same congruences. For large u , let $n_e = (N - t_e)/p_e$ and form

$$W(N) = \sum_e w_e \sigma_k(p_e) T_{n_e}.$$

If α_k is rational, every T_{n_e} is eventually integral:

$$W(A + Qu) \in \mathbb{Z} \quad \text{for all sufficiently large } u.$$

3. Only the top denominator power remains

After expansion and dilation, a typical contribution is

$$\left\{ \begin{matrix} i \\ j \end{matrix} \right\} \sum_e w_e p_e^{i+1} \frac{\sigma_k(N + r_e(j))}{(N + r_e(j))^{i+1}}, \quad 0 \leq i, j \leq k.$$

For every $i < k$:

- If $j < k$, hold all coordinates except e_j fixed. The shifted value is constant in e_j ; the polynomial degree is $i + 1 \leq k$. Its weighted sum is zero.
- If $j = k$, the Stirling coefficient $\left\{ \begin{matrix} i \\ k \end{matrix} \right\}$ is zero.

All rows $i < k$ vanish identically. Only $i = k$ survives.

This is an exact finite-sum identity, not an estimate.

3. Name what survives before taking any limits

Define the normalized divisor sum and the remaining coefficients:

$$\rho_k(m) = \frac{\sigma_k(m)}{m^k}, \quad c_{e,j} = w_e p_e^{k+1} \left\{ \begin{matrix} k \\ j \end{matrix} \right\}.$$

Then cancellation gives

$$W(N) = S(N) + E(N), \quad S(N) := \sum_e \sum_{j=0}^k c_{e,j} \frac{\rho_k(N + r_e(j))}{N + r_e(j)}.$$

Here $E(N) = \sum_e w_e \sigma_k(p_e) E_{n_e}$ is a signed error. For each vertex, $N = p_e(n_e + 1) - r_e(0)$, so

$$NE_{n_e} = p_e(n_e + 1)E_{n_e} - r_e(0)E_{n_e} \longrightarrow 0.$$

The family is fixed and finite; therefore $NE(N) \rightarrow 0$.

First limit: the weighted integer is eventually zero

All limits here are along the same progression $N = A + Qu$. From the divisor bound,

$$0 \leq \frac{\rho_k(m)}{m} \leq \frac{64}{\sqrt{m}} \longrightarrow 0.$$

There are only finitely many fixed coefficients and shifts, so

$$S(N) \longrightarrow 0, \quad E(N) \longrightarrow 0.$$

Thus $W(N) = S(N) + E(N) \rightarrow 0$.

Use integrality at exactly this point

Eventually $W(N) \in \mathbb{Z}$ and $|W(N)| < 1$. Hence $W(N) = 0$, and consequently $S(N) = -E(N)$.

Second limit: remove the remaining denominators

Define the **survivor**, with exactly the same coefficients and shifts:

$$R(N) := \sum_e \sum_{j=0}^k c_{e,j} \rho_k(N + r_e(j)).$$

For each fixed term, write $r = r_e(j)$ and $c = c_{e,j}$. Then

$$N \frac{c \rho_k(N + r)}{N + r} - c \rho_k(N + r) = -\frac{r c \rho_k(N + r)}{N + r} \rightarrow 0.$$

Therefore $NS(N) - R(N) \rightarrow 0$. But the preceding slide gives $NS(N) = -NE(N) \rightarrow 0$.

The consequence of rationality

$\alpha_k \in \mathbb{Q}$ would force $R(A + Qu) \rightarrow 0$.

One shift has a coefficient that cannot cancel away

The shift

$$r_* = (k+1)B$$

occurs *only* at the zero vertex $e = 0$ and the last position $j = k$.

- For $j < k$, $r_e(j) \leq k(B + DG) = (k+1)B - 1$.
- For $j = k$, $r_e(k) = (k+1)B + D \sum_{a < k} (k-a)e_a(k+2)^a$. All added coefficients are positive, so equality requires $e = 0$.

Its coefficient is

$$c_* = w_0 B^{k+1} \begin{Bmatrix} k \\ k \end{Bmatrix} = B^{k+1} > 0, \quad w_0 = (-1)^{k(k+1)} = 1.$$

Other shifts may repeat; this one does not.

The survivor is concrete when $k = 1$

Recall $p_e = 5, 7, 9$, $t_e = 0, 2, 4$, and $w_e = 1, -2, 1$. The progression has modulus $Q = (5 \cdot 7 \cdot 9)^2 = 99225$.

$$W(N) = 6T_{N/5} - 16T_{(N-2)/7} + 13T_{(N-4)/9}.$$

The common first shift cancels because $5 - 14 + 9 = 0$. The second shifts have coefficients 25, -98 , 81, giving

$$R(N) = 25\rho_1(N + 10) - 98\rho_1(N + 12) + 81\rho_1(N + 14).$$

Rationality would force this expression to tend to 0 on the progression.

The shift 10 stands alone, with positive coefficient 25.

4. Divisor sums become divisor densities

Pair complementary divisors to rewrite

$$\rho_k(m) = \frac{\sigma_k(m)}{m^k} = \sum_{d|m} \frac{1}{d^k} \quad (m > 0).$$

Along a progression $Qn + a$ with $Q, a > 0$, how often does a divisor occur?

Example $Qn + a = 6n + 3$	$d = 2$	$d = 3$	$d = 5$	$d = 9$
Density of $d \mid 6n + 3$	0	1	1/5	1/3

In general, with $g = \gcd(d, Q)$,

$$\text{density of } d \mid Qn + a = \begin{cases} g/d, & g \mid a, \\ 0, & g \nmid a. \end{cases}$$

The compatible congruence has one solution class modulo d/g .

4. The actual progression mean is explicit

For fixed $k \geq 1$ and $Q, a > 0$,

$$\frac{1}{X} \sum_{n=0}^{X-1} \rho_k(Qn + a) \longrightarrow \mu_Q(a) := \sum_{\substack{d \geq 1 \\ \gcd(d, Q) | a}} \frac{\gcd(d, Q)}{d^{k+1}} \geq 1.$$

Why may we sum the individual divisor densities? Counted values $Qn + a$ are distinct positive multiples of d , so for $X \geq 1$,

$$\frac{1}{d^k} \frac{\#\{n < X : d \mid Qn + a\}}{X} \leq \frac{QX + a}{X d^{k+1}} \leq \frac{Q + a}{d^{k+1}}.$$

This bound is summable in d . Dominated convergence applies.

This includes $k = 1$: the majorant is a multiple of d^{-2} .

4. A fresh prime changes the mean by one factor

Take a prime $L \nmid Q$ and refine the progression to $Q(Ln + v) + a$. Its starting point $a' = Qv + a$ is congruent to a modulo Q .

$$\mu_{QL}(a') = \mu_Q(a) \left(1 - L^{-(k+1)} + \mathbf{1}_{[L|a']} L^{-k} \right).$$

The indicator $\mathbf{1}_{[L|a']}$ is 1 if $L \mid a'$, and 0 otherwise.

- The terms with $L \nmid d$ contribute $\mu_Q(a)(1 - L^{-(k+1)})$.
- Write the other terms as $d = Ld'$. They contribute $\mu_Q(a)L^{-k}$ if $L \mid a'$, and zero otherwise.

There is a common contribution, plus an extra term when the prime divides the offset.

4. Choose one residue to miss and one to isolate

Write $i = (e, j)$, $r_i = r_e(j)$ and $c_i = c_{e,j}$, so $R(N) = \sum_i c_i \rho_k(N + r_i)$. Choose a prime $L > \max(Q, \max_i r_i)$. Since $\gcd(Q, L) = 1$, choose:

Miss every shift

$$Qv_0 + A \equiv 0 \pmod{L}$$

$$Qv_0 + A + r_i \equiv r_i \pmod{L}$$

Never zero: $0 < r_i < L$.

Hit only the isolated shift

$$Qv_1 + A + r_* \equiv 0 \pmod{L}$$

$$Qv_1 + A + r_i \equiv r_i - r_* \pmod{L}$$

Zero exactly when $r_i = r_*$.

These give two subprogressions of $N = A + Qu$. There is exactly one term at r_* , with coefficient $c_* > 0$.

4. The two averages cannot both vanish

If $R(A + Qu) \rightarrow 0$, its mean on *each* positive-step subprogression is zero. But the prime-refinement formula gives the two means

$$m_0 = (1 - L^{-(k+1)}) \sum_i c_i \mu_Q(A + r_i),$$
$$m_1 = m_0 + c_* \mu_Q(A + r_*) L^{-k}.$$

Every offset $A + r_i$ is positive, so the mean theorem applies. Their difference satisfies

$$m_1 - m_0 = c_* \mu_Q(A + r_*) L^{-k} > 0,$$

because $c_* > 0$, $\mu_Q(A + r_*) \geq 1$, and $L^{-k} > 0$.

Contradiction

Both means would be zero. Hence α_k is irrational for every $k \geq 1$.

Degree zero returns to the short tail argument

For $k = 0$, the first denominator term and the same error estimate give

$$T_n = \frac{\sigma_0(n+1)}{n+1} + E_n.$$

The divisor count is positive and at most $64\sqrt{n+1}$, so

$$0 < T_n \leq \frac{64}{\sqrt{n+1}} + O(n^{-3/2}) \longrightarrow 0.$$

Rationality would make these tails integers for all sufficiently large n . That is impossible once $0 < T_n < 1$.

The separate zero-degree argument completes every natural exponent.

The formal artifact proves the original series

$$\forall k \in \mathbb{N}, \quad \text{Irrational} \left(\sum_{n=0}^{\infty} \frac{\text{ArithmeticFunction.sigma}(k, n)}{n!} \right).$$

The $n = 0$ term is zero. The statement audit also checks the expanded divisor sum, summability, and the equivalent series starting at $n = 1$.

- Lean 4.33.1, pinned Mathlib; full build and a fresh trust-zero build.
- Statement and axiom audits; fresh replay using Lean's kernel.
- Exactly `propext`, `Classical.choice`, and `Quot.sound`.

Proof module: **812 substantive lines; 1,090 total lines**. The one-line root import adds one to each count.

Source: Lean source and reproduction commands.

Three ingredients resolve the contradiction

1. **Factorials turn rationality into integrality.** Every sufficiently late scaled tail would be an integer.
2. **A fixed dilation grid creates exact cancellation.** The weighted integer tends to zero, leaving one identifiable survivor.
3. **Progression means detect that survivor.** A fresh prime separates two averages that rationality would force to agree.

Conclusion

For every natural k , $\sum_{n \geq 1} \sigma_k(n)/n!$ is irrational.

Appendix: the finite Stirling bound used in Lean

Put

$$D_{j+1}(x) = \frac{1}{x(x-1)\cdots(x-j)}, \quad P_{j,L}(x) = \sum_{i=0}^{L-1} \begin{Bmatrix} i \\ j \end{Bmatrix} x^{-(i+1)}, \quad F_{j,L}(x) = D_{j+1}(x) - P_{j,L}(x).$$

The exact recurrence is

$$F_{j+1,L+1}(x) = \frac{F_{j,L}(x) + (j+1)F_{j+1,L}(x)}{x}.$$

For $j+1 \leq H \leq x$, induction on L gives

$$0 \leq F_{j,L}(x) \leq H^L/x^{L+1}.$$

At $L=0$, $0 < D_{j+1}(x) \leq 1/x$. In the induction step, the weighted sum of two errors bounded by u is at most $(j+2)u \leq Hu$. The $j=0$ error is zero after the first order.

This proves the truncated expansion directly, with an explicit error.

Appendix: scale each remainder before taking limits

For $x \geq n + 1$, the divisor bound gives

$$\sigma_k(x) \leq 64x^k \sqrt{x} \leq \frac{64x^{k+1}}{\sqrt{n+1}}.$$

For a retained term, take $x = n + j + 1$ and $n \geq k + 1$. Scale first:

$$\begin{aligned} 0 \leq (n+1)\sigma_k(x)F_{j,k+1}(x) &\leq x \frac{64x^{k+1}}{\sqrt{n+1}} \frac{(k+1)^{k+1}}{x^{k+2}} \\ &= \frac{64(k+1)^{k+1}}{\sqrt{n+1}} \rightarrow 0. \end{aligned}$$

There are only $k + 1$ retained terms, so their scaled errors sum to a null sequence. For the omitted tail O_n , factorial denominators give

$$0 \leq (n+1)O_n \leq \frac{128(k+1)^{k+1}}{\sqrt{n+1}} \rightarrow 0.$$

Adding the two limits proves $(n+1)E_n \rightarrow 0$ directly.

Appendix: a compact notation guide

$k \geq 1$	fixed divisor-power exponent in the grid argument
$e \in \{0, \dots, k+1\}^k$	one vertex of the finite grid
p_e, t_e	multiplier and offset for that vertex
$r_e(j) = (j+1)p_e - t_e$	shift of the $(j+1)$ -st tail term
$w_e = \prod_{a < k} (-1)^{k+1-e_a} \binom{k+1}{e_a}$	integer finite-difference weight
$A, Q = \prod_e p_e^2$	CRT starting point and progression modulus
$n_e = (N - t_e)/p_e$	tail index, for large $N = A + Qu$
$c_{e,j} = w_e p_e^{k+1} \{^k_j\}$	coefficient after cancellation
$r_* = (k+1)B, c_* = B^{k+1}$	the isolated shift and its coefficient
L, v_0, v_1	fresh prime and the two separating residues

Appendix: four quantities with different roles

$$T_n = n! \sum_{m>n} \frac{\sigma_k(m)}{m!}$$

one scaled tail,

$$W(N) = \sum_e w_e \sigma_k(p_e) T_{n_e}$$

the weighted integer,

$$S(N) = \sum_e \sum_{j=0}^k \frac{c_{e,j} \rho_k(N + r_e(j))}{N + r_e(j)}$$

the surviving divided sum,

$$R(N) = \sum_e \sum_{j=0}^k c_{e,j} \rho_k(N + r_e(j))$$

the raw survivor.

The normalized divisor sum is $\rho_k(m) = \sigma_k(m)/m^k$. The logical chain is

$$W = S + o(1/N), \quad W \in \mathbb{Z}, \quad W \rightarrow 0 \implies W = 0 \text{ eventually} \implies NS \rightarrow 0 \implies R \rightarrow 0.$$

The last implication uses $NS - R \rightarrow 0$, not the false identity $NS = R$.

Sources and the complete proof

- **Formal proof and full mathematical exposition**

Tokengrinder, *Erdős 252* (2026). github.com/tokengr1nder/Erdos252

PROOF.pdf contains all 92 named proof results and their proofs, 38 definitions, and the six statement-audit results.

- **Historical context**

K. Pratt, *The irrationality of a divisor function series of Erdős and Kac* (2022), introduction.

arXiv:2209.11124.

- J. B. Friedlander, F. Luca, M. Stoiciu, *On the irrationality of a divisor function series*, *Integers* 7 (2007), A31.

Original paper.

- **Problem catalogue**

T. F. Bloom, Erdős Problem 252.